

AMENDMENT No. 1 MAY 2026
TO
IS 3025 (Part 44) : 2023 METHODS OF SAMPLING AND TEST (PHYSICAL AND
CHEMICAL) FOR WATER AND WASTEWATER
PART 44 BIOCHEMICAL OXYGEN DEMAND (BOD)
(Second Revision)

(Page 3, clause 9.5, line 1) — Substitute the following for the existing:

‘The following four types of dilutions are:’

(Page 3, clause 9.5.3) — Substitute the following for the existing:

9.5.3.1 Fill a 1 litre volumetric flask with approximately 300 ml dilution water (9.1) as prepared. Add the specific volume of sample in it as per dilution technique given in Table 1. If the dilution water is not feeded with nutrients during aeration, then add 1 ml each of phosphate buffer solution (7.1), magnesium sulphate solution (7.2), calcium chloride solution (7.3) and ferric chloride solution (7.4) one by one in the volumetric flask.

9.5.3.2 If the sample requires seeding, add the requisite volume of seed which meets the GGA check of 198 mg/l $\pm$ 30.5 mg/l as mentioned in Table 2 and dilute to the mark with the dilution water by siphoning from the dilution water container. Mix well. Rinse three BOD bottles with the diluted sample and fill up these bottles with the diluted sample by siphoning from the volumetric flask. Stopper the bottles immediately after removing the air bubbles. Repeat the same process for other dilutions of the sample. For preparing a seed control, fill up a 1 litre volumetric flask partly with the dilution water, add 20 ml of seed (2 percent of freshly collected or stabilised settled sewage) in it and 1 ml each of the nutrients and fill up to the mark with the dilution water. The volume of seed added may vary depending on the quality of seed used.

9.5.3.3 For preparing a seed control, fill up a 1 litre volumetric flask partly with the dilution water, add 20 ml of seed (2 percent of freshly collected or stabilised settled sewage) in it and 1 ml each of the nutrients and fill up to the mark with the dilution water. The volume of seed added may vary depending on the quality of seed used.’

(Page 4, Table 2, Title) — Substitute the following for the existing:

‘Table 2 Method for Direct Pipetting the Samples into BOD Bottles if Large Number of Samples are Handled (Clause 9.5.3.1)’

(Page 5, clause 9.6) — Substitute the following for the existing:

‘9.6 Use of Seed Control

Seed control refers to the determination of the biochemical oxygen demand (BOD) of the seed suspension in the same manner as for any other sample. Ideally, prepare three different dilutions of seed such that smallest quantity depletes 2.0 mg/l or more of dissolved oxygen (DO) and the largest quantity leaves 1.0 mg/l or more of dissolved oxygen (DO) residual after 5 d of incubation. The dissolved oxygen (DO) uptake per milliliter of seed is calculated by dividing the dissolved oxygen (DO) depletion by the volume of seed in milliliters for each seed control bottle with a 2.0 mg/l depletion and more than 1.0 mg/l minimum residual dissolved oxygen (DO), and averaging the results. Seed dilutions showing widely varying depletions per milliliter of seed ($\pm$ 30 percent) suggest the presence of toxic substances or large particulates in the seed suspension; check or change the seed source.’